

Decision-making guide

Ordering

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1. Simulation platform introduction

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes market, decision phases, the Decision checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

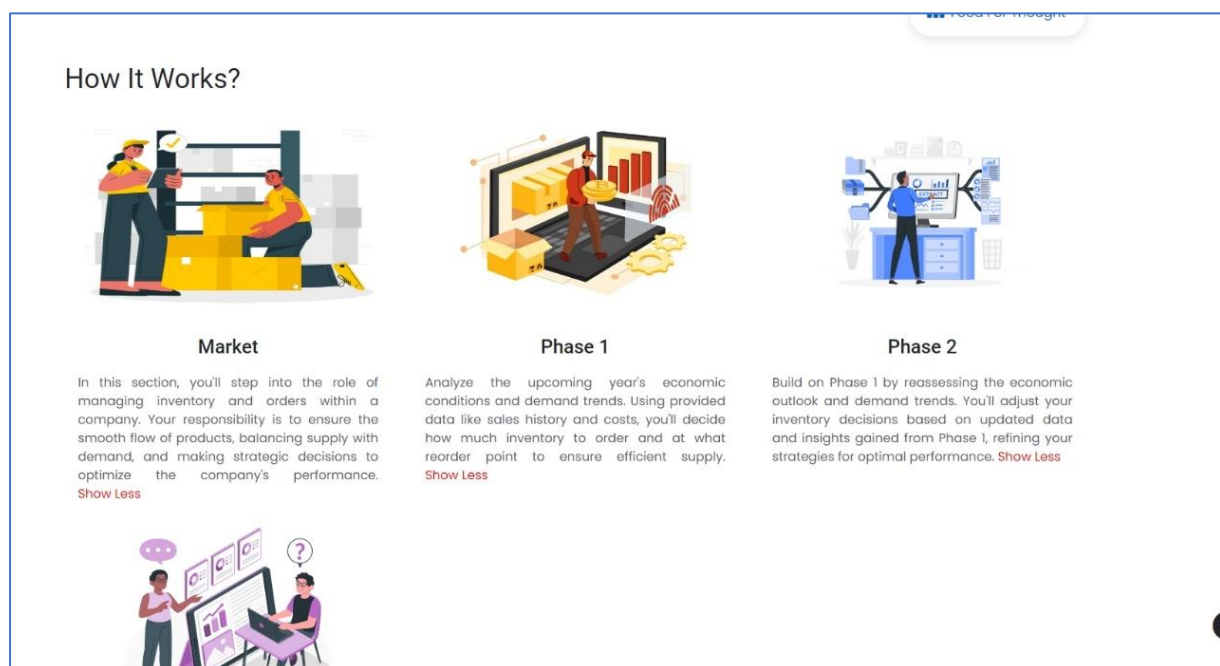


Figure 1 Game Arena

1.2 Readings

This section provides essential materials necessary for participants to actively engage in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the simulation. This video presents insights into the industry, ongoing trends, obstacles, and the diverse array of decisions required.

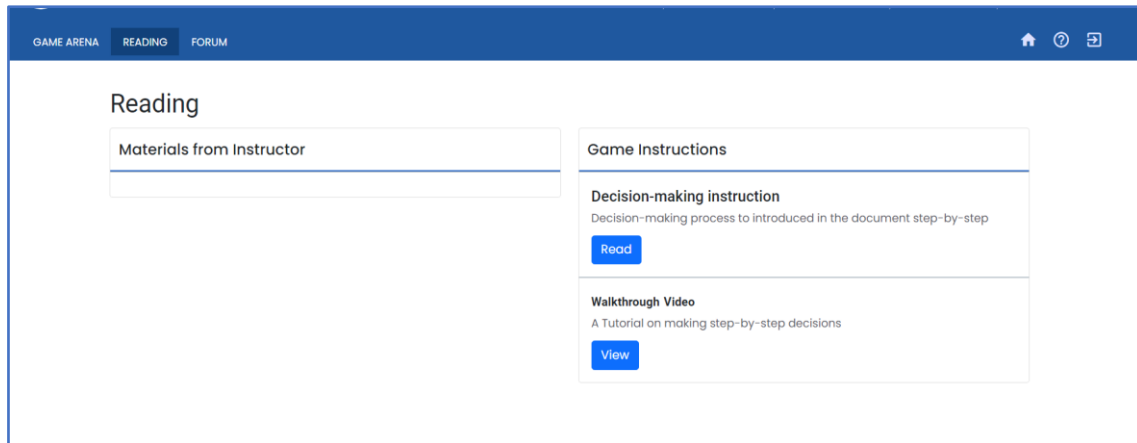


Figure 2 The Reading Section

1.3 Forum

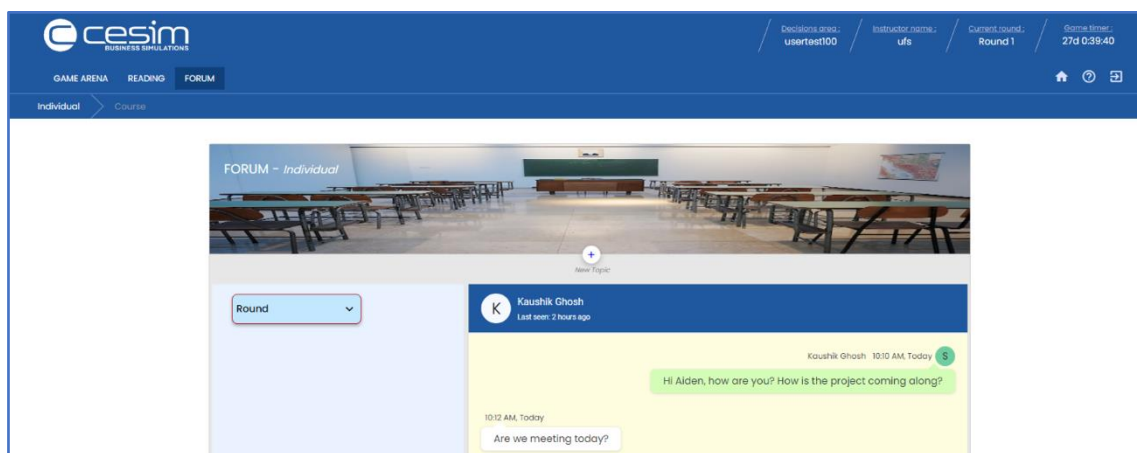


Figure 3 The Forum Section

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

Dextrous Art, an Indian start-up founded by a group of artists, specializes in unique and affordable artistic home décor items. With aspirations to capture a share of the booming Indian home décor market, projected to reach ₹121.97 billion by 2030, the company has faced challenges in planter sales despite a pull strategy. Issues like high stockouts and ineffective inventory management have hindered growth.

To address this, the research team developed a patented real-time forecasting system to optimize inventory levels and reorder decisions. The management is considering shifting to a push strategy, leveraging social media campaigns and special deals to create demand and boost sales. The participant role involves utilizing the forecasting system to enhance service levels while minimizing costs.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

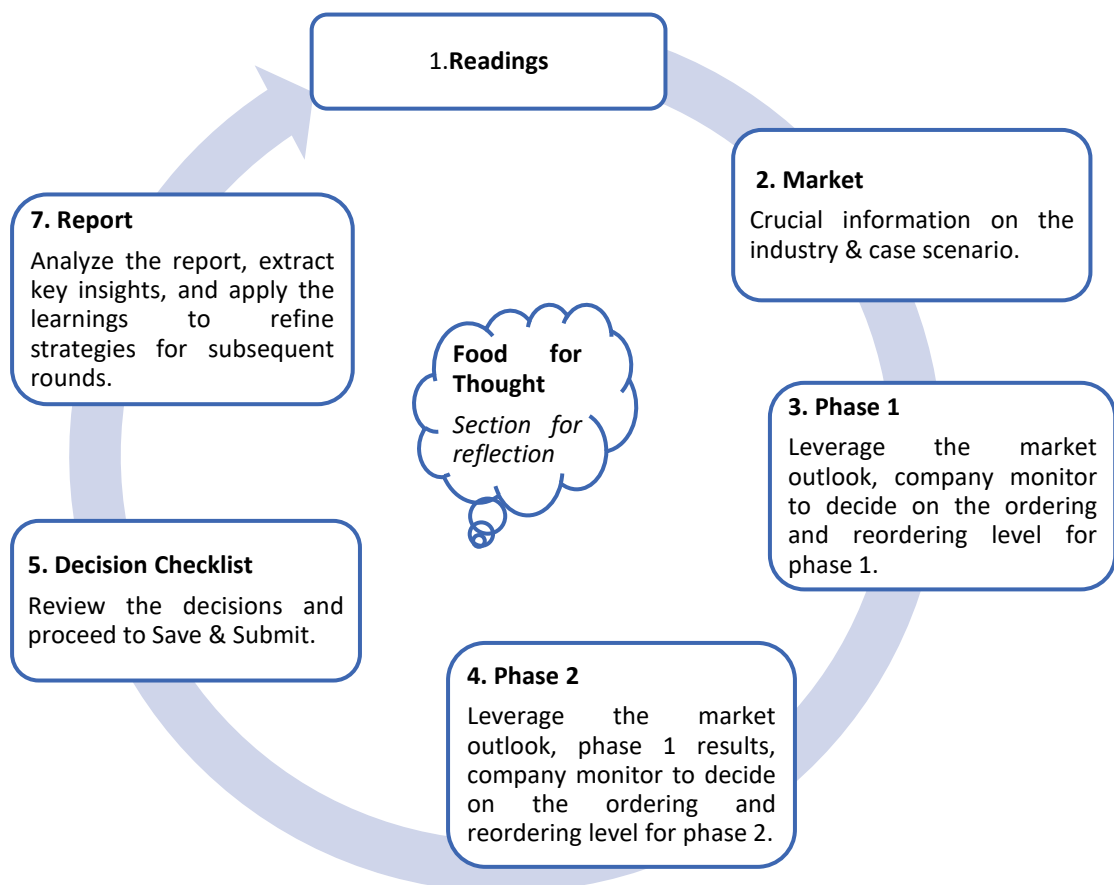


Figure 4 The Sequence of Decision-Making Process

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

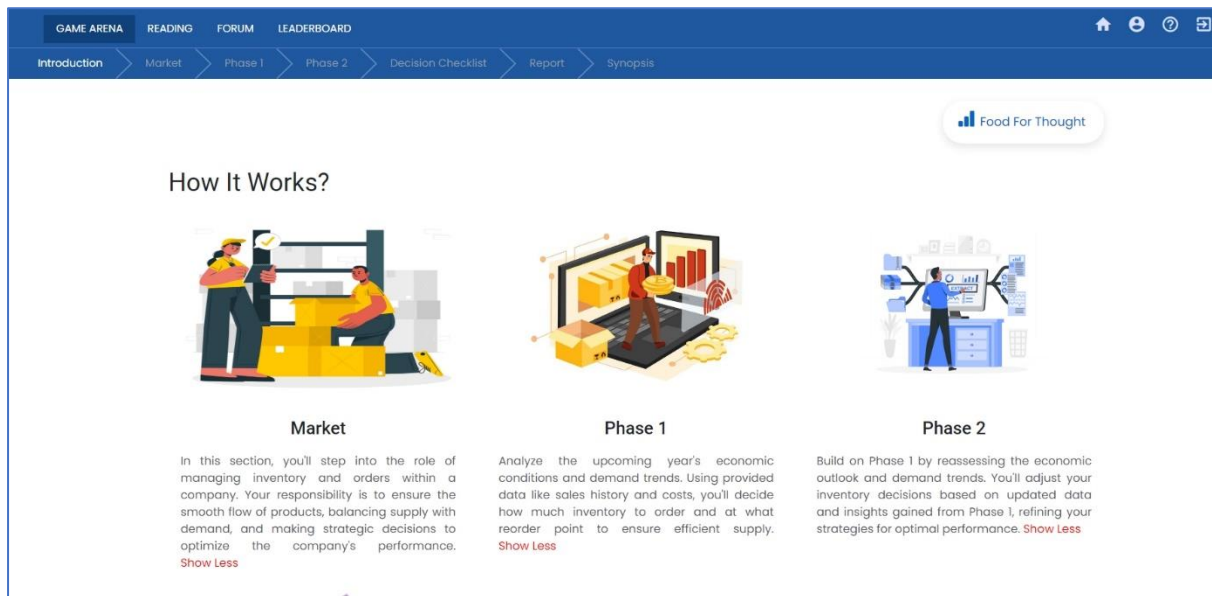


Figure 5 The Game Arena

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making.

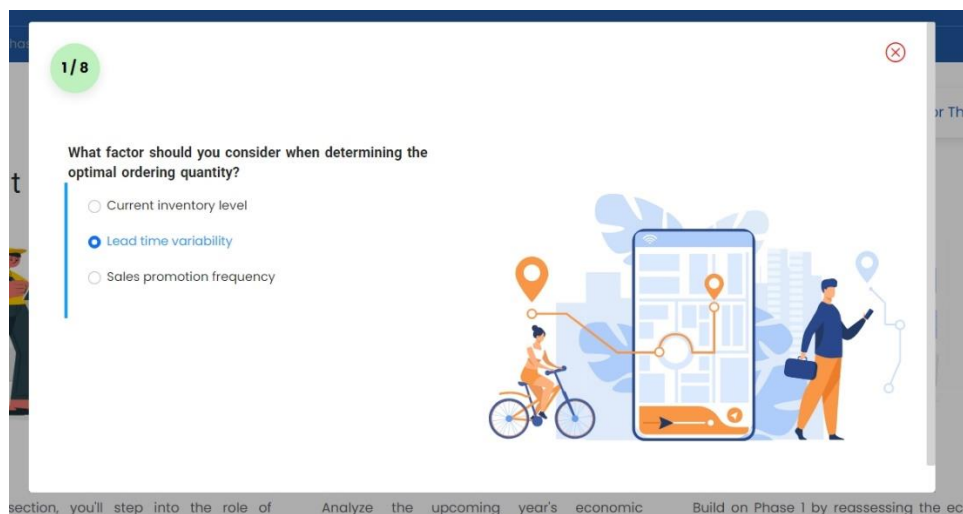


Figure 6 The Food for Thought Section

2.3 Market

Prior to commencing the decision-making process, it is essential to carefully examine the "Market" section. This section offers vital information on case scenario, industry overview and the key parameters to consider when making decisions in each tab.



Figure 7 Market Tab

2.4 Phase 1

Phase 1: Decision-Making

Decision-making begins in Phase 1, where the Outlook presents the economic scenario. Participants are provided with essential data for planning and tasked with developing efficient reorder and order quantity strategies.

To support informed decision-making, participants analyse inventory data and metrics available in the Company Monitor. The company's research team has extrapolated inventory data for a 52-week period, which is accessible in this section. Key metrics, including ordering costs, holding costs, stockout costs, and backordering costs, are also provided to take informed decision.

Company Monitor: Key Values

Mean: The mean demand represents the average demand, serving as the central point around which demand fluctuates.

- **Standard Deviation:** Standard deviation measures the variability or dispersion of demand from the mean. For example, a standard deviation of 13 indicates that demand typically varies by 13 units from the mean.
- **Variance:** Variance, the square of the standard deviation, is another measure of demand variability. It highlights how spread out the demand values are.
- **Median:** The median is the middle value in the demand distribution, dividing the dataset into two equal halves, with 50% of values above and 50% below it.
- **Mode:** The mode represents the most frequently occurring demand value in the dataset.
- **25th Percentile:** The 25th percentile (Q1) indicates that 25% of the demand values fall below this point. It can be a critical benchmark for determining the minimum stock level to avoid frequent stockouts.
- **50th Percentile:** Also known as the median, the 50th percentile shows the point where half of the demand values are below and half are above.
- **75th Percentile:** The 75th percentile (Q3) reflects the demand value below which 75% of the observations fall, providing insight into moderately high demand levels.
- **95th Percentile:** The 95th percentile demonstrates that 95% of the time, demand will be below this value, making it a key reference for planning for extreme demand scenarios.

Applying the Data to Inventory Management

1. Reorder Level:

- The reorder level is the stock level at which new inventory should be ordered to avoid disruptions during the replenishment lead time.
- While the mean demand provides a baseline for estimating typical lead-time demand, accounting for variability (e.g., using the standard deviation or percentiles) ensures better preparedness for demand fluctuations.
- Reorder levels could be set using higher percentiles (e.g., 75th or 95th) to ensure sufficient stock during periods of higher-than-expected demand.

2. Safety Stock:

- Safety stock addresses variability in demand and prevents stockouts during demand surges.
- If demand variability is high (as indicated by the standard deviation), safety stock levels should be determined using higher percentiles, such as the 95th percentile.
- This ensures that inventory can accommodate unexpected spikes in demand.

3. Order Quantity:

- Order quantity is typically estimated using average demand and lead time, but incorporating safety stock and reorder levels into calculations ensures inventory sufficiency.
- By considering variability in demand, businesses can strike a balance between minimizing holding costs and preventing stockouts.

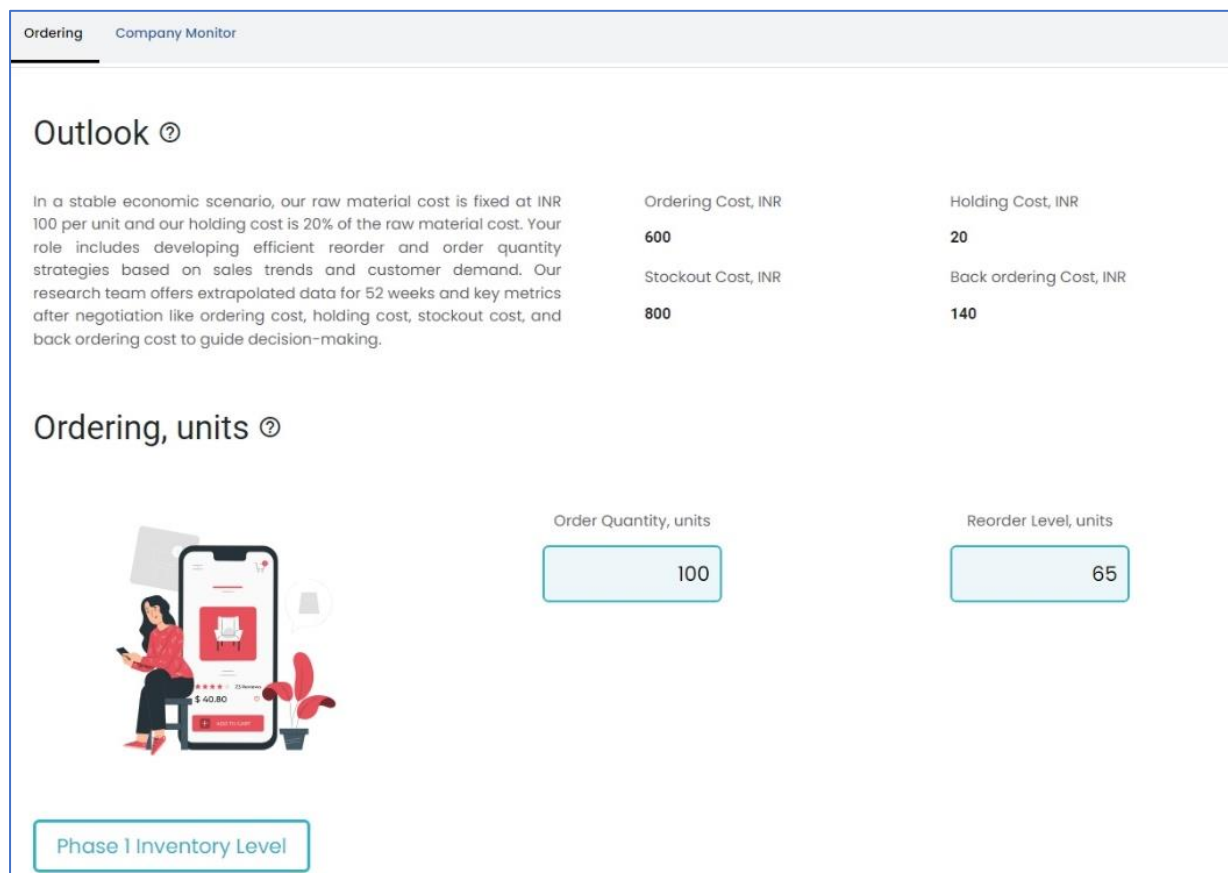
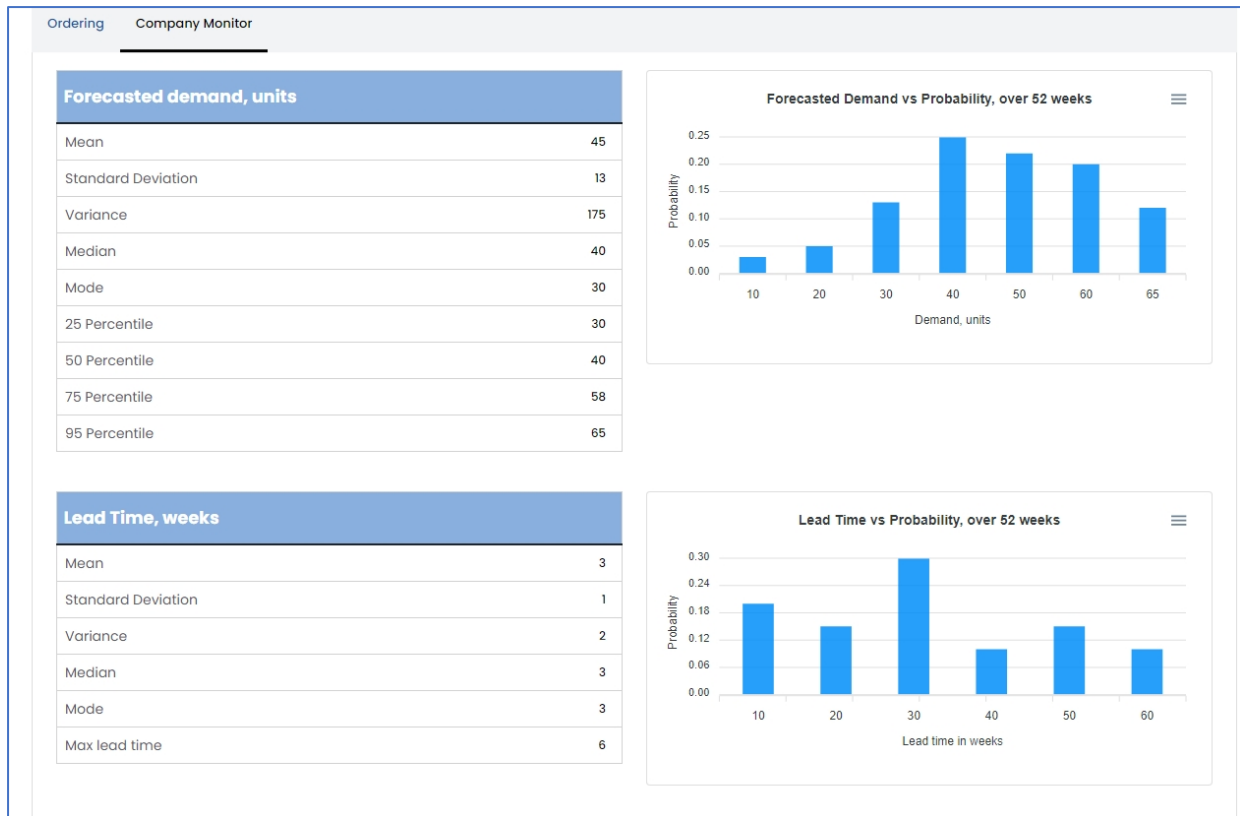


Figure 8 Phase 1 Decision Area



2.6 Decision checklist

The Decision Checklist helps participants review and adjust their decisions before the deadline. Once decisions are confirmed, they can submit them using the "Submit" option and choose [save and proceed](#). It's important to remember that once decisions are submitted and saved, they cannot be changed for that round. This ensures that decisions are carefully considered before finalization, promoting clarity and accountability in the decision-making process.

Decision Checklist

Round 1 ▼

User

Phase 1

Order Quantity, units	280
Reorder Level, units	135

Phase 2

Order Quantity, units	311
Reorder Level, units	230

Submit

Help
ⓧ

"You are submitting your decisions as the final decision. Do you want to save and proceed with reports?"

Close
Save & Proceed

Figure 10 The Decision Checklist

2.7 Report

The report section provides the details on the participants' critical thinking abilities in each aspect. It demonstrates how their thought process influenced the outcomes, providing insights into the impact of their decision-making skills.

This section provides feedback on their thought process regarding the "Food for Thought" sections across four metrics.

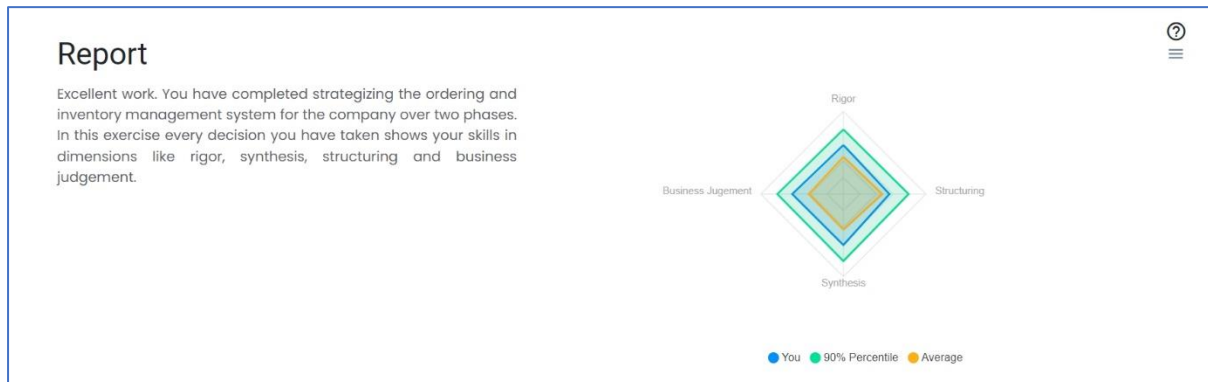


Figure 11 The Report Section

- **Rigour:** Consistency throughout the decision-making and recommendation.
- **Structuring:** Logical approach of gathering and analysing information
- **Synthesis:** Approach towards interpreting information and making informed decisions
- **Business Judgement:** Ability to evaluate facts, and consider risks and possible implications when making decisions.

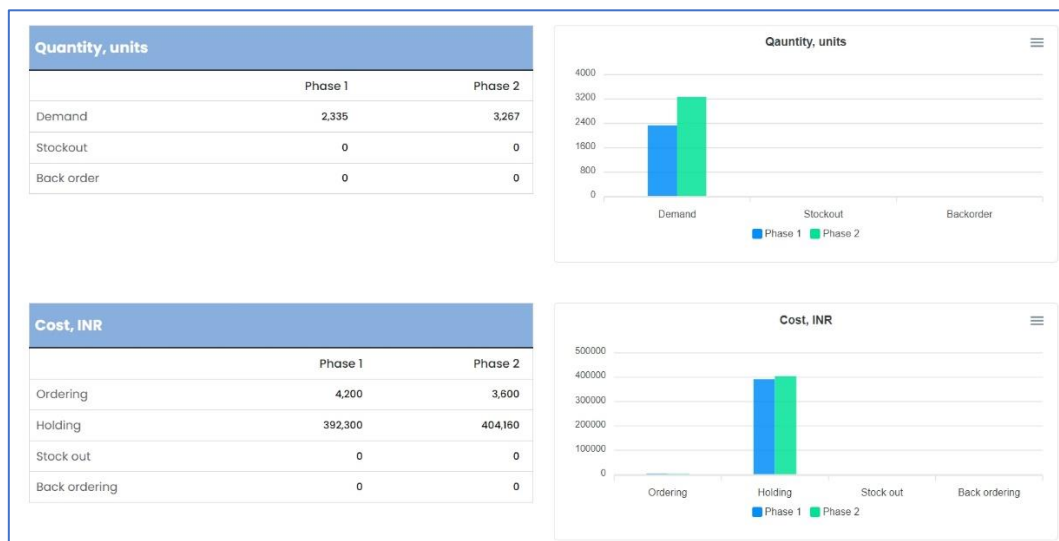


Figure 12 The Report Section

Here in this section of report, the actual demand, stock out and backorder happened for two phases are available. The related cost associated with is also visible.

- **Actual Demand:** This refers to the total quantity of products requested or required by customers during a specific period.
- **Stock Out:** This occurs when inventory levels are insufficient to meet the demand, leading to a situation where products are unavailable for customers.
- **Backorder:** This happens when orders cannot be fulfilled immediately due to stock shortages but are scheduled for later delivery.

The KPI section highlights how effectively the participants improved the KPIs and the total costs incurred to achieve these improvements. Striving for a higher service level at a lower cost represents an ideal scenario for participants to aim for.

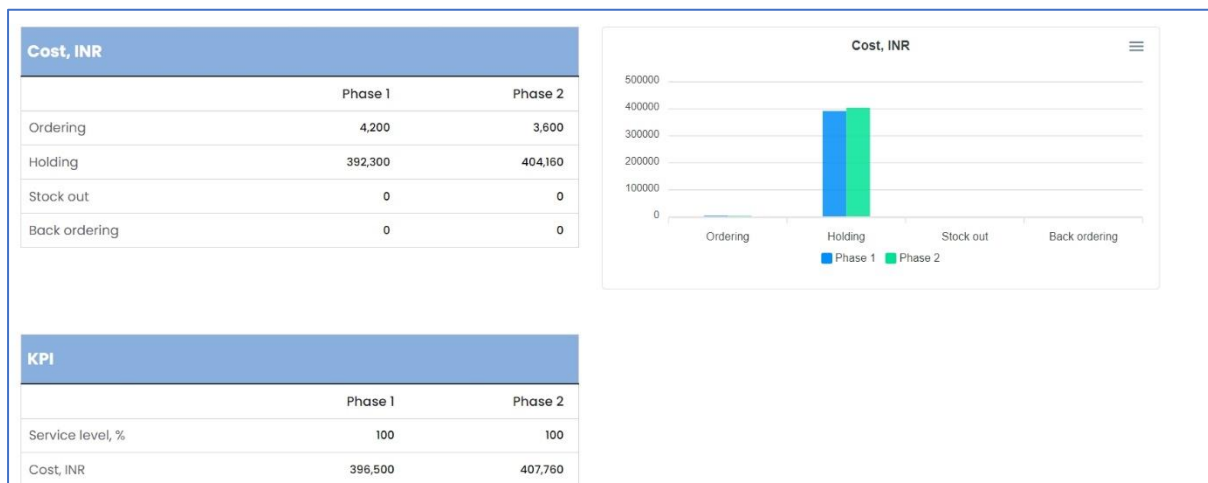


Figure 13 The Report Section showing the KPI

2.8 Synopsis

The synopsis section offers a concise summary of key decision areas and outcomes, highlighting critical insights including the KPI achieved and strategic directions taken during the simulation, facilitating comprehensive understanding and future planning.

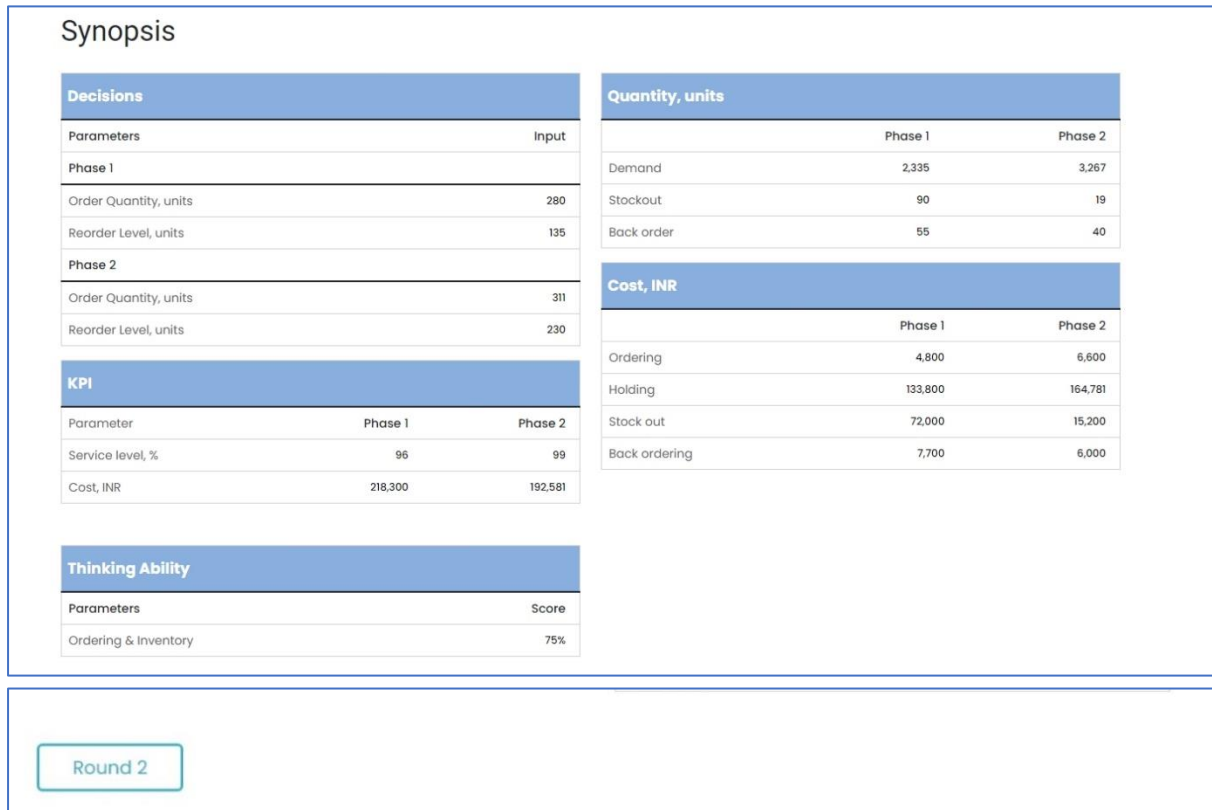


Figure 14 The Synopsis Section

To proceed to the next iteration, follow these steps:

1. Navigate to the end of the synopsis page to locate the link for the next round.
2. Click on the 'Round' to advance to the next round.
3. You will be redirected to the home page.
4. Click on the card to begin the next round.